Q6. Numeric Center

A numeric center is a number that separates in a consecutive and positive integer number list (starting at one) in two groups of consecutive and positive integer numbers, in which their sum is the same. For example, the numeric center of the list {1, 2, 3, 4, 5, 6, 7, 8} is number 6, which produces two lists of consecutive and positive integer numbers (i.e. {1, 2, 3, 4, 5} and {7, 8}) in which their sums are the same (in this case 15). As another example, the numeric center of the list {1, 2, 3, 4, …, 49} is 35, which produces two lists {1, 2, 3, 4, …, 34} and {36, 37, 38, 39, …, 49}, and the sum of each list is equal to 595.

Input

The input consists of 6 lines. There is one test case for each line. Each test case contains a positive integer number n $(1 \leq n \leq 100000000)$.

Output

For each test case, you have to print one line containing the numeric center of the list {1, 2, 3, 4, …, n}. Print “null” if there is no numeric center in the test case.

Sample Input

127
8
57121
39782
49
6748

Sample Output

null
6
40391
null
35
null